Sagnik Ghatak

Munich, Germany

Robotics, Autonomy & AI Research Engineer

+49 1755538219

ghatak.sagnik5@gmail.com

sghatak5

sghatak5

Professional Summary

Robotics and AI Research Engineer with hands-on experience in autonomous systems, multi-robot coordination, and machine learning. Skilled in full-stack robotics development - from ROS2-based navigation and manipulation pipelines to federated reinforcement learning and LLM-driven task planning. Proven ability to bridge research and deployment across collaborative robots (KUKA, UR10e, TurtleBot), simulation environments (IsaacSim, Gazebo), and MLOps infrastructure. Passionate about building intelligent, safe, and scalable robotic systems.

Professional Experience

AImotion Bavaria — Ingolstadt, Germany
Wissenschaftlicher Mitarbeiter — 04/2026 - 09/2026

- Developed a client-server communication framework for remote robot motion control, implementing a **Java-based REST** client on the KUKA LBR iiwa 7 R800 to enable real-time dispatch of heterogeneous motion commands from an external server.
- Designed a modular AI interface for collaborative robot systems integrating **LLM-based task planning** and a **PPR-extended OWL semantic environment model**, converting natural language instructions into executable robot motion sequences.

Cognizant Technology Solutions — Kolkata, India
Data Engineer — 10/2021 - 09/2023

- Designed and optimised **ETL pipelines** in Python to ingest, clean, and transform large-scale datasets, building reliable data infrastructure for downstream analytics and ML model training.
- Automated **data preprocessing and workflow orchestration** using Python and UNIX scripting, reducing manual intervention by 60% and accelerating feature-ready dataset delivery for time-sensitive tasks.

Student Experience

AImotion Bavaria — Ingolstadt, Germany
Studentische Hilfskraft - Robotics — 03/2025 - 03/2026

- Deployed a full autonomous navigation stack on TurtleBot4 using **ROS2 and Nav2**, configuring SLAM, costmaps, and recovery behaviours for real-time obstacle avoidance and goal-directed navigation.
- Integrated **MoveIt2** on a Braccio robotic arm to plan and execute collision-free manipulation trajectories, bridging simulation-validated motion plans to physical hardware.
- Developed **Python scripts** to publish goal positions directly to Nav2 and MoveIt2 action servers, enabling repeatable automated task execution without manual RViz input.
- Engineered a **pick-and-place pipeline** on a **UR10e** cobot using **Polyscope** and **OnRobot Eye** for object detection, pose estimation, and closed-loop grasp execution.

UniCredit S.p.A — Munich, Germany
Working Student - Data Science / MLOps — 07/2024 - 02/2026

- Containerised ML models using **Docker** and orchestrated deployments on **Kubernetes**, maintaining CI/CD pipelines via **GitHub Actions** to automate training, testing, and rollout across staging and production environments.

Skills

Robotics & Autonomy: ROS2, Nav2, MoveIt2, SLAM, Motion Planning, Manipulation
AI & Machine Learning: Deep Learning, Reinforcement Learning, Federated Learning, LLM-based Task Planning, SKRL
Simulation: Gazebo, IsaacSim, Foxclove
MLOps & DevOps: Docker, Kubernetes, GitHub, CI/CD
Programming: Python, C++, Java, MATLAB, UNIX Scripting
Tools & Platforms: Git, REST APIs

Education

Technische Hochschule Ingolstadt Ingolstadt, Germany
M.Eng. AI Engineering of Autonomous Systems 10/2023 - 03/2026
Thesis: A Simulation Architecture for Privacy-Preserving Multi-Robot Learning

Maulana Abul Kalam Azad University of Technology Kolkata, India
B.Tech Electrical Engineering 07/2017 - 06/2021
Thesis: Speed Control of DC Motor Using Fuzzy Logic

Projects

A Simulation Architecture for Privacy-Preserving Multi-Robot Learning 2025
Master Thesis · IsaacSim, Reinforcement Learning, Federated Learning

- Architected a **Federated Deep Reinforcement Learning (FDRL)** framework integrating IsaacSim, SKRL, and Flower for privacy-preserving multi-agent navigation training across distributed robot teams.
- Implemented and validated **MAPPO** baselines achieving **100% system success rate** and **0% collision rate** in coordinated TurtleBot3 navigation, establishing performance upper bounds for federated comparison.
- Demonstrated that federated MAPPO matches centralised coordination while maintaining data sovereignty, validating decentralised learning viability for real-world multi-robot deployments.

RoboCup Rescue - Unitree Go2 Arm Integration Ongoing
Robotics Team, THI · MoveIt2, Foxglove, ROS2, Manipulation

- Established a **DDS-to-ROS2 communication bridge** for the Unitree D1 robotic arm, enabling real-time command dispatch and sensor feedback within the ROS2-based rescue autonomy stack.
- Built a **simulation and visualisation environment** in **Foxglove Studio** for live telemetry monitoring, joint state inspection, and motion debugging during field deployment.
- Implementing a **MoveIt2 motion planning pipeline** for the D1 arm, enabling collision-aware trajectory planning for autonomous manipulation in RoboCup Rescue scenarios.

Formula Student Autonomous - State Estimation & Path Planning 03/2024 - 08/2024
Schanzer Racing Electric e.V · ROS2, Python, EKF

- Implemented a **dual Extended Kalman Filter (EKF)** fusing IMU and wheel odometry for high-precision vehicle state estimation under dynamic racing conditions.
- Programmed **path planning algorithms** for real-time trajectory generation around track landmarks in a Formula Student Driverless context.

Local Path Planner for Evasive Manoeuvres of Automated Vehicles 2024
Research Project · Python, Vehicle Dynamics, Control Systems

- Developed a local path planner for robust evasive manoeuvres using a **bicycle model** and **PID controller**, improving obstacle avoidance in dynamic environments.

Echo Bot - Autonomous SLAM Robot 2024
ROS2, Python, Gazebo

- Developed an autonomous robot with full **SLAM capabilities** using ROS2, simulated in Gazebo for mapping and navigation validation.

Languages

English: C1 **German:** A2